Corollary 8.4.1: For any tuple of skew shapes ν with $|\nu| = \ell$, we have the following raising operator style formula for ∇ applied to the associated LLT polynomial:

$$(\omega\nabla G_\nu)(z_1,\dots,z_\ell) = (-1)^{p(\nu)}(qt)^{p(\nu)+n'(\mathfrak{T}(\nu))}\cdot q^{A(\nu)}\,\sigma\left(\frac{z^{\lambda+(1,\dots,1)}\prod_{\alpha\in R_{qt}}(1-qtz^\alpha)}{\prod_{\alpha\in R_q}(1-qz^\alpha)\prod_{\alpha\in R_t}(1-tz^\alpha)}\right)_{pol}$$

where $R_q, R_t, R_{qt}, \lambda$ are defined in Def. 7.1.2, $A(\nu) = \#$ attracting pairs in ν.

Note: We did not use $H_\nu^{1,0}$ which also works here by Theorem 7.3.2, but we cannot apply Prop 3.5.1 which requires $n=1$.

* The different between $H_\nu^{1,1}$ and $H_\nu^{1,0}$ is λ. In $H_\nu^{1,1}$, $\vec{b}=(1,1,\dots,1)$. In $H_\nu^{1,0}$, $\vec{b}=(0,0,\dots,0)$. So if $H_\nu^{1,0} = H(R_q,R_t,R_{qt},\lambda)$, then $H_\nu^{1,1} = H(R_q,R_t,R_{qt},\underbrace{\lambda+(1,1,\dots,1)}_{\text{This is } \lambda \text{ in §8.2, which is } \hat{\lambda}+\vec{b}(m,n)})$

defined in Def 7.1 which is $\hat{\lambda}$ in §8.2

Proof: Note that $G_\nu = \mathrm{cub}((-1)^{p(\nu)}(qt)^{p(\nu)+n'(\mathfrak{T}(\nu))}\cdot q^{A(\nu)}H_\nu^{1,1})$ by Theorem 8.3.1. by Prop. 3.5.1, we have

$$(\omega\nabla G_\nu)(z_1,\dots,z_\ell) = (-1)^{p(\nu)}(qt)^{p(\nu)+n'(\mathfrak{T}(\nu))}\cdot q^{A(\nu)}\cdot\sigma\left(\frac{z^{\lambda+(1,1,\dots,1)}\prod_{\alpha\in R_{qt}}(1-qtz^\alpha)}{\prod_{\alpha\in R_q}(1-qz^\alpha)\prod_{\alpha\in R_t}(1-tz^\alpha)}\right)_{pol}.$$

□

More generally, we have

Corollary 8.4.2: For any tuple of skew shapes ν with $\ell = m|\nu|$, we have the following raising operator style formula for ∇^m applied to the associated LLT polynomial:

$$(\omega\nabla^m G_\nu)(z_1,\dots,z_\ell) = (-1)^{p(\nu)}(qt)^{p(\nu)+m\,n'(\mathfrak{T}(\nu))}\cdot q^{A(\nu)}\,\sigma\left(\frac{z^{\lambda}\prod_{\alpha\in R_{qt}}(1-qtz^\alpha)}{\prod_{\alpha\in R_q}(1-qz^\alpha)\prod_{\alpha\in R_t}(1-tz^\alpha)}\right)_{pol}$$

where $R_q, R_t, R_{qt}, \lambda$ are defined in Def. 8.2.1 with $n=1$ and the m-stretching $\nu = (m, (c,c,c,\dots,c))$ of ν, $\mathcal{A} = (m\cdot\#$ attracking pairs in ν with the same content$)+(\#$ attracking pairs with different contents$)$

Proof: Note that $G_\nu = \mathrm{cub}((-1)^{p(\nu^m)}(qt)^{p(\nu^m)+n'(\mathfrak{T}(\nu^m))}q^{\sum_{(i,j)\in A(\nu)} a_{ij}}H_{\nu^m}^{m,1})$ by Theorem 8.3.1. Also $p(\nu^m)=p(\nu)$ and $n'(\mathfrak{T}(\nu^m)) = m\cdot n'(\mathfrak{T}(\nu))$ by Remark 8.3.2 (i), and by Remark 8.3.2 (ii), $\sum_{(i,j)\in A(\nu)} a_{ij} = (m\cdot\#$ attracting pairs in ν with the same content$)+(\#$ attracting pairs with different contents$)$. Hence $\mathcal{A} = \sum_{(i,j)\in A(\nu)} a_{ij}$.

$$\therefore\ G_\nu = \mathrm{cub}((-1)^{p(\nu)}(qt)^{p(\nu)+mn'(\mathfrak{T})}q^{\mathcal{A}}H_{\nu^m}^{m,1}).$$

By Prop. 3.5.1,

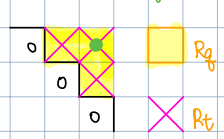
$$(\omega\nabla^m G_\nu)(z_1,\dots,z_\ell) = (-1)^{p(\nu)}(qt)^{p(\nu)+m\,n'(\mathfrak{T}(\nu))}\cdot q^{A(\nu)}\,\sigma\left(\frac{z^{\lambda}\prod_{\alpha\in R_{qt}}(1-qtz^\alpha)}{\prod_{\alpha\in R_q}(1-qz^\alpha)\prod_{\alpha\in R_t}(1-tz^\alpha)}\right)_{pol}.$$

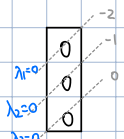
□

Remark: Corollary 8.4.2 holds for any m-stretching $\nu(m,\vec{b})$ of m (i.e. $\vec{b}$ may not be a constant sequence), but now with $\mathcal{A} = \sum_{(i,j)\in A(\nu)} a_{ij}$ (b/c Remark 8.3.2 (ii) only works for constant sequences $\vec{b}$).

e.g. $\nu = ((1,1,1))$, then $R_q = R_t = R_{+} = \{\alpha_{12},\alpha_{13},\alpha_{23}\}$ ($R_t \setminus R_q = \phi$ b/c each diagonal has only 1 cell. $R_q \setminus R_t = \phi$ b/c there is only one shape, i.e. no attracting cells)

$R_{qt} = \{\alpha_{13}\}$ (b/c $R_t \setminus R_{qt} = \{\alpha_{12},\alpha_{23}\}$ b/c $\nu(1),\nu(2)$ and $\nu(2),\nu(3)$ are on adjacent diagonals)





$$H_\nu^{1,0} = \sum_{w\in S_3} w\left(\frac{1-qt\frac{z_3}{z_2}}{\prod_{1\le i<j\le 3}(1-\frac{z_i}{z_j})(1-q\frac{z_i}{z_j})(1-t\frac{z_i}{z_j})}\right) \quad\text{and}\quad H_\nu^{1,1} = \sum_{w\in S_3} w\left(\frac{z^{(0,0,0)+(1,1,1)}(1-qt\frac{z_3}{z_2})}{\prod_{1\le i<j\le 3}(1-\frac{z_i}{z_j})(1-q\frac{z_i}{z_j})(1-t\frac{z_i}{z_j})}\right)$$

$$= \sum_{w\in S_3} w\left(\frac{z_1z_2z_3\,(1-qt\frac{z_3}{z_2})}{\prod_{1\le i<j\le 3}(1-\frac{z_i}{z_j})(1-q\frac{z_i}{z_j})(1-t\frac{z_i}{z_j})}\right)$$

By Corollary 8.4.1,

$\lambda = (0,0,0)$

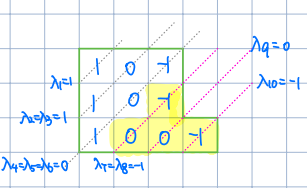
$p(\nu) = n(\mathfrak{T}(\nu)) = 0$, $A(\nu) = 0$

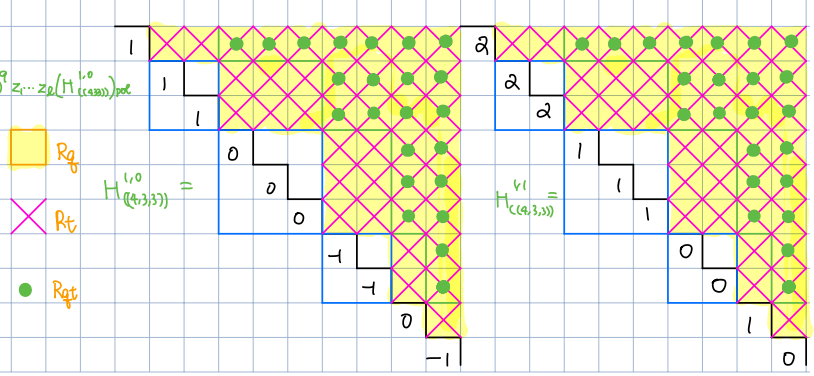
$$(\omega\nabla e_3)(z_1,z_2,z_3) = \omega\nabla G_{(1,1,1)} = \sigma\left(\frac{z_1z_2z_3\,(1-qt\frac{z_1}{z_3})}{\prod_{1\le i<j\le 3}(1-q\frac{z_i}{z_j})(1-t\frac{z_i}{z_j})}\right)_{pol} = s_{111} + (q+t+q^2+qt+t^2)s_{21} + (qt+q^3+q^2t+qt^2+t^3)s_3.$$

e.g. $\nu = ((4,3,3))$, $G_\nu = s_{433}$, $H_{((4,3,3))}^{1,1} = z_1\cdots z_9 H_{((4,3,3))}^{1,0}$. In §7.2, we computed $p((4,3,3)) = 4$, $n'(\mathfrak{T}(\nu)) = 5$, $A(\nu) = 0$

By Corollary 8.4.1,

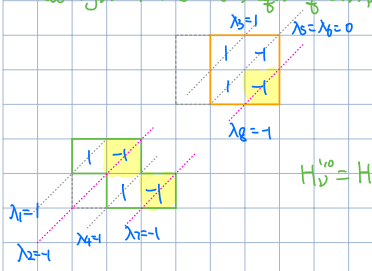
$$(\omega\nabla s_{433})(z_1,\dots,z_9) = (-1)^4(qt)^{4+5}q^0(H_{((4,3,3))}^{1,1})_{pol} = (qt)^9 z_1\cdots z_9(H_{((4,3,3))}^{1,0})_{pol}$$



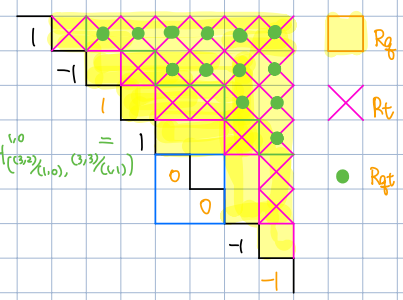
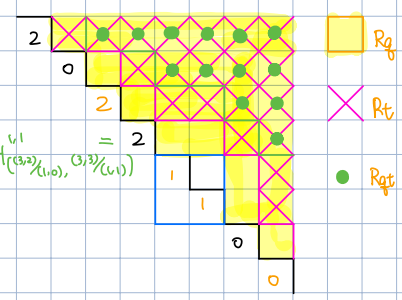


e.g. $\nu = ({}^{(3,2)}/_{(1,0)}, {}^{(3,3)}/_{(1,1)})$. In §7.2, we computed $p(\nu)=3$, $n'(\sigma(\nu))=1$, $A(\nu)=7$.

$$(\omega \nabla G_\nu)(z_1,\dots,z_\ell) = (-1)^3 (qt)^{3+1} q^7 (H^{1,1}_\nu)_{pol} = -q^{11} t^4 (z_1 z_2 \dots z_\ell \cdot H^{1,0}_\nu)_{pol}$$



$H^{1,0}_\nu = H^{1,0}_{({}^{(3,2)}/_{(1,0)}, {}^{(3,3)}/_{(1,1)})} =$



$H^{1,1}_{({}^{(3,2)}/_{(1,0)}, {}^{(3,3)}/_{(1,1)})} =$